

OPTIMUS PRIME / EXPEDITION SKILLS FOR HUMANOID MOUNTAINEERING

THE MOUNTAIN AMPLIFIES FAILURE

A SLIP BECOMES A FALL.

A FALL BECOMES A RESCUE.

G1 must use tools, recover, and know when to stop.



FOUR SKILLS FOR EXTREME TERRAIN.

ONE EXPEDITION SYSTEM / CURRICULUM-TRAINED POLICIES



01 SELF-ARREST

STOP THE
SLIDE



02 FALL RECOVERY

GET BACK
UP



03 CONTROLLED RAPPEL

DESCEND WITH
CONTROL



04 TEAM LIFT

MOVE HEAVIER
LOADS TOGETHER

HOW SINGLE-ROBOT SKILLS LEARN

MUJOCO / PPO / HIERARCHICAL CONTROL

SELF-ARREST PPO

125-D OBSERVATION →

2 × 256 TANH →

14 ARM RESIDUALS

100 Hz · ≈200k actor+critic params

FIXED-LINE RECOVERY

FALL + ROPE STATE →

2 × 256 TANH →

4-ACTION HANDOFF

50 Hz · pretrained 29-DoF WBC prior

REWARD SHAPING / PHYSICS MUST EXPLAIN SUCCESS

01 TOOL CONTACT

rigid pick contact
blade 22–42° · snow load

02 BODY CONTROL

chest-down pose · grip
wrists stay ahead of torso

03 TERMINAL GATES

no passive stop
stable stand earns success

CURRICULUM

0.25–0.75 m/s

START EASY



→ 5.0 m/s

SPEED



±40° / ±1.5 m/s

HEADING + CROSS-SLOPE



HARD ANCHORS

ROLL + ADVERSARIAL FALLS



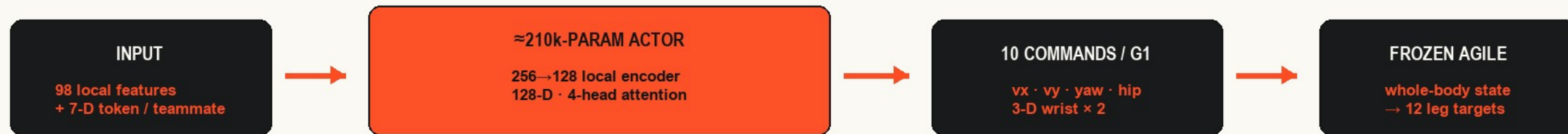
9/9 + 60/60

FROZEN SELECTION

HOW THE TEAM POLICY LEARNS

ISAAC SIM / ISAAC LAB / SHARED MAPPO / CURRICULUM

SHARED ATTENTION ACTOR / SAME WEIGHTS ON EVERY G1



TRAINING-ONLY CENTRAL CRITIC

TEAM STATE + PAYLOAD MASS

768 → 512 → 256

ACTOR: LOCAL + TOKENS ONLY
AT INFERENCE

MAPPO: 24-step rollouts · 5 epochs · γ .99 · GAE .95 · clip .2 · lr 3e-4

REWARD SHAPING / SHARED TEAM OBJECTIVE

+ TRACK

+ LEVEL

+ HEADING

+ LOAD BALANCE

+ UPRIGHT

+ LIFT PROGRESS

- EXTENSION / RATE / FALL

CURRICULUM

LIFT + LEVEL

0-24k steps

ADD CARRY + TURN

24k→150k

MASS 8→18 kg

by 180k

2 / 3 / 5 G1

shared actor

LIVEKIT CONNECTS THE EXPEDITION

ONE ROOM / TWO EXPLICIT CONTROL PLANES

ROBOT STATE PLANE / LIVEKIT DATA TRACKS



OPERATOR VOICE PLANE / LIVEKIT AGENTS + INFERENCE



VOICE SUPERVISES · LOCAL POLICIES CONTROL MOTION

WHAT WE MEASURED

FIXED TESTS / PHYSICAL OUTCOMES



SELF-ARREST

9 / 9 SCENARIOS

60 / 60 unseen falls arrested



FALL RECOVERY

RECOVERS TO STAND

then continues uphill



RAPPEL

2.00 m DESCENT

7 / 10 randomized starts passed



TEAM LIFT

50 / 50 SHARE

overload or imbalance → abort

MOVEMENT / ACTION / THINKING

SIM-TO-REAL: OPEN CHALLENGES

SCALING TO THE HIMALAYAS / THE C.L.I.M.B. FRONTIER

C

CONTACT

Snow & Ice Dynamics

- Deformable snow crust
- Non-linear friction drops
- Slush & sliding scree
- Dynamic soft terrain

L

LIGHTING

Optics & Sensors

- Flat-light whiteout
- Extreme snow glare
- Harsh cliff shadows
- Precipitation LiDAR clutter

I

INTER-COMMS

Field Mesh Network

- No cloud LiveKit at altitude
- Ad-hoc local wireless mesh
- Packet drops >150 ms deadline
- Relative UWB sensing

M

MECHANICAL

Ropes & Tethers

- Dynamic rope elasticity
- Variable carabiner friction
- Dynamic line bounce & sag
- Line snag & twist hazards

B

BALANCE

Unassisted Authority

- Withdraw sim balance assists
- Whole-body 43-DoF torque
- Sub-zero battery drop
- Thermal actuator limits

BRIDGING THE GAP FROM SIMULATION TO HARSH ALPINE DEPLOYMENT